Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/03/2010
PI Name:	Dunn-Rankin, Derek Ph.D.		
Project Title:	Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-3975	**Congressional District:**	48
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:	Flight definition. NOTE: Extended to 9/30/2011 per NSSC info (9/2009) NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weinberg, Felix (Imperial College, London, England) Yuan, Zeng-Guang (National Center for Space Exploration Research, NASA Glenn Research Center)		
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Task Description:	This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving the performance of energy conversion systems. Electric fields have been shown to affect flame stability and soot formation; while as a sensing device, the flames electrical response can describe flame characteristics. In the past year, we have continued electric field testing on a coflow burner and jet diffusion burner for both methane and ethylene, which are proposed for the ISS experiment along with fundamental studies involving partially-admixed air/fuel jet flames. Specific activities over the year included: 1) Participating in the Requirements Definition Review to finalize requirements for our experiment and assure they are within CIR capabilities. 2) Irvine/NASA Glenn testing included, measuring the flame's electrical response and voltage-current characteristic under various conditions, and reduced gravity testing at the NASA Glenn 2.2-s Drop Tower. 3) Experiments and numerical simulations of CO, H2, and ion formation in flames with varying levels of aeration. 4) Experiments designed to measure the ion current of a flame as it is progressively quenched. 5) Blackbody calibration of cameras for ground-based studies and development of algorithms to process color-filtered images from hardware expected for the ISS experiment for soot pyrometry. Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways. First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way. Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Several of the applications we explore have potential uses in earth gravity. Understanding fundamental electrical properties of flames have contributions to earth-based combustion, particularly as regards soot formation and control. Ion current is also a very useful diagnostic tool and sensor for combustion, capable of reflecting flame character and behavior for next generation power generation systems such as homogeneous charge compression ignition engines. Finally, there are important potential applications in combustion synthesis of nanoparticles where ion formation pathways are important.
	UCI Coflow Burner -- Experimental work was completed using a coflow burner. The burner consists of two stainless steel plenums threaded together. Fuel is mixed, if desired, with nitrogen in the bottom plenum and passed through a stainless steel tube (2.13 mm ID). The flow at the exit is fully developed. Air is introduced in the top chamber, passing through beads, and a honeycomb mesh to create a uniform flow at the exit. A variable high potential supply creates an electric field between the burner and mesh electrode. Field strength between the electrodes is not linear due to the flame's space charge; however, the field can be approximated as E = V/d, where V is the applied potential and d is the distance between electrodes. This approximation is used in describing the experiments. Ion current is computed using Ohm's law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber protects the burner from room currents. The measurement circuitry includes provisions to protect the data acquisition hardware from accidental discharges: damping resistors, diodes, and an isolation amplifier. NASA Glenn Coflow Burner -- At the NASA Glenn Research Center, the burner apparatus includes a cylindrical test chamber, flow system, and data acquisition/control system built into a NASA 2.2 sec. Drop Tower rig. The coflowing burner is placed coaxially inside the test chamber. A stainless-steel fuel tube is coaxially positioned inside a Pyrex converging nozzle, through which air flows. Gas flow rates were measured by mass flow meters. To dampen out spurious fluctuations, present only in microgravity environments, a flow orifice was placed downstream of the mass flow meters. Recently, the mass flow meters were found responsible for the fluctuations and subsequently replaced with

Task Progress:

a critical flow orifice as the rig's sole monitoring device. A hot-wire igniter is used to ignite the fuel. To provide a uniform flow, two layers of honeycomb plates, two stainless steel mesh screens atop the honeycomb, and two additional honeycomb plates are placed in the base of the nozzle. The burner does not have a honeycomb mesh in contact with the fuel tube at the jet exit, creating a deformed field converging toward the jet tube in contrast to the quasi-parallel electric field lines around the base of the jet when a honeycomb is present. This, however, is a minor issue when considering the electrical nature of the flame, which deforms the electric field lines due to the flame's space charge.

Methane flame studies at NASA Glenn -- Experiments at the NASA 2.2 Second Drop Tower are aimed at removing gravity's influence on a flame under the influence of an electric field. The similar influence of buoyant convection and the ion-driven wind make it difficult to decouple the effects in a 1g environment. The results will help isolate the relevant physics of a buoyant free, electrically manipulated flame and facilitate tailoring the test matrix for the ISS mission. By removing the influence of large coflow velocities and buoyancy, videos and ion current measurements can show the complex interaction between ionization, flame shape, and ion-driven wind. In a µg environment, flames take on a spherical shape owing to the lack of a preferential transport direction of hot products (buoyancy). When a field is applied, the resulting ion wind drives away combustion products, much like buoyancy, and entrains air to the reaction zone. Under field between -0.6 and -1kV/cm, the flame will mimic the structure of a 1g flame. Along with images of the flame, ion current measurements were taken both in 1g and µg environments. Regardless of gravity level both V-I curves have similar features. The ion current exhibits minor differences when changing gravity levels with an upward directed field (negative applied potential); however, by changing polarity in µg, the flame is quenched on the burner jet, weakening the flame and the ion current. Once an electric field is applied to a flame, the flame's settling time, is dependent on the apparent gravity field acting on the flame. In a 1g environment, the flame takes on an egg shape, resulting from the buoyant acceleration of hot products from the reaction zone. Any additional velocity resulting from the ion wind creates a small change in the flame promptly (~10ms). In a µg environment, buoyancy is not a factor leaving the ion wind as the flame's only external influence. The experimental results show the transition between two applied voltages (-1kV to -2kV). With a net upward wind, fresh oxidizer is brought to the reaction zone, increasing the reaction rate and the ion current, further increasing the ion wind strength. The ion current reaches a maximum at 600ms, after which the ion current begins to decrease as a result of the increase in soot. Soot is an electron emitter and has been shown to interfere with ion current measurements. Recently, efforts have included a study to remove the contribution of soot radiation in chemiluminescent CH* measurements, a precursor species to ion formation.

Partial aeration studies -- Our Imperial College collaborative activity is examining the effects of partial aeration on the structure of flames and its relation to saturation ion current, CO and H2 formation, and understanding the hazardous conditions created by poorly tuned industrial boilers. Results in the literature showed that saturation ion current would scale with the amount of carbon present in a diffusion flame; however, when flames are fully premixed, saturation ion current is a function of temperature. Between these established regimes, there exists a region largely unexplored, where air is premixed in concentrations below the rich flammability limit. In our studies, appreciable concentrations of carbon monoxide appear within the flame cone, particularly for burners with a spatial scale of the order of a diffusion flame thickness, which raised concerns regarding the potential hazard that if by partially quenching the flame, dangerous levels of CO were released into the ambient atmosphere. Hence, the object of the current study was to investigate, both experimentally and by numerical simulation, the formation and spatial distribution of CO and H2 under the aerated small flame conditions relevant to burners in domestic boiler systems and then to determine the effect of various quenching surfaces on CO escape into the surroundings.

The burner system is a simple fuel jet. The fuel flow rate was fixed for each series and a progressively increasing proportion of air was admixed. The appearance and shape of the flames does not vary appreciably, in spite of the substantial increase in total flow velocity. This is in accord with classical diffusion flame theory, which predicts that these quantities depend on the volumetric flow rate of fuel alone, even in the presence of other gases. This applies until the mixture becomes flammable. Once the air admixture crosses the 85% boundary (all concentrations are expressed as % by volume), an inner cone forms, gradually detaches and shrinks, reaching a minimum around the stoichiometric composition and then grows again.

Sampling was carried out at a grid of points across an axial plane by means of a quartz microprobe. Since visual observation and flame photographs showed the flame base to overhang the burner by several millimetres, sampling was extended to the edge of flame luminosity. The aspirated gas was analysed for its CO and H2 content by carbon monoxide and hydrogen detectors, respectively. A traverse across the burner diameter using a chromel-alumel (K-type) thermocouple with a 30 mm diameter wire was carried out at a height of 2 mm above the burner rim, as a function of air admixture (and hence flow velocity).

Quenching -- The main sequence of measurements was obtained using an apparatus comprised of a flame trap matrix above the burner mouth. The flame trap matrix was wound, using alternately plane and corrugated stainless steel strips, into a cylindrical plug. To vary the quenching surface, grids of different metals could be placed immediately upstream of the plug. Using this system, the average CO emitted by an uncooled burner is around 2.6ml/min. When other quenching surfaces were interposed ahead of the matrix, the expanded aluminium laminate led to the largest CO emissions once again. Results included values in excess of 3 ml/min, even for a flame with no air addition. At 75% air this rises to 4 ml/min. The highest CO emission rate measured was about 4ml/min. This is close to 10% of the CH4 fed into the burner and clearly poses a potentially serious hazard. Expressing this in terms of the maximum carbon monoxide release for a practical boiler yields about 0.177 L/min/kW. For the 30kW recommended for a 1 bedroom flat this would exceed 5 litres per minute.

Numerical simulation -- Numerical simulations were performed using GRI-Mech Version 3.0 and the chemical kinetics program, CRESLAF, included in the CHEMKIN package, to predict gas-phase temperatures, velocities and concentrations. The model reduces the Navier-Stokes equations by making a boundary-layer assumption that a principal flow direction exists, where convection is the primary transport mechanism and diffusion is neglected axially (radial diffusion is retained). After discretizing the relevant equations, the solver, DASSL, marches axially by minimizing the formulated residual at every grid point at a given height, before moving downstream.

Although there have been many significant numerical simulation studies of small coflow diffusion flames, having to provide insight for the specific geometry, air admixing conditions, and quenching experiments of our study demanded that we develop customized simulations to match our experimental conditions. Using the same inlet velocities as in the experiment, we produced concentration profiles analogous to experimentally measured values. Results show that although regions of maximum CO concentrations are near the flame centerline; the maxima in CO flux are shifted towards regions of larger radius, as shown clearly in the figure. We were unable to simulate the quenching effects of the

flame trap matrix; however, summing the differential volume flux at a given height above the burner would represent the amount of CO released, were the downstream kinetics frozen. Scaling the volume flow rate to 300K shows trends comparable to our experimental results, with the maximum CO flux occurring at 4mm.

The numerical simulations resemble the experimental results. The CO concentration for 75% air again exceeds that for pure fuel, though the difference is greater in the simulation than in the experiment; also the peak in the core of a vortex around the axis in the former case is not reproduced. The contour appears more mushroom shaped - though a single point could transform it from one to the other geometry. We attribute the difference to the assumption of a parabolic flow profile in the simulation, as against the flow pattern above the thick - walled burner port. Preliminary experimental studies using the cork borer burner at higher flow velocities show a highly arched biconvex contour, rather than a closed vortex.

Electrical Properties of Quenched Flames -- In addition to our efforts at Imperial College, parallel studies are underway at UC Irvine, studying the electrical properties of partially quenched flames. Our work at Imperial College has illuminated the potential dangers of improperly turned boilers and the high concentration of carbon monoxide and hydrogen potentially emitted when quenched. This portion of our research is aimed at understanding the electrical behavior of a plate incrementally lowered toward a flame and measuring the corresponding ion current. The study is a proof of concept to explore the potential of detecting flame impingement electrically without visual inspection. Preliminary studies involve plates of varying metals placed above an axisymmetric jet flame with a small potential difference between the two surfaces (~3V). Ion current is measured after a few minutes to ensure thermal equilibrium. Significant amplification is necessary due to the small voltages applied. As the plate is lowered toward the flame there is a gradual increase in ion current, an increase is attributed to the increase in electric field strength. Ion current sharply increases at distances below 3 cm, finally saturating the A/D module of the data acquisition hardware at 1cm above the jet exit, at which point, the flame sheet is interrupted by the quenching plate. From this point the burner acts as a low resistance pathway between both electrodes allowing current to flow freely. Further lowering the plate toward the jet exit, the ion current decreases as the flame temperature and ion production falls. The next phase of this work will involve actively cooling the plate and emissions sampling.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
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